Quadratic equations

Subject: Mathematics | Level: Class 10 | Curriculum: Sindh Board
Prepared by: SHS Virtual Academy

1. Topic Overview and Learning Objectives

Quadratic equations form the backbone of secondary school algebra in the Sindh Board curriculum. An equation of the form ax^2 + bx + c = 0, where a is not equal to zero, is defined as a quadratic equation. Understanding these equations is fundamental because they appear in almost every branch of higher mathematics, physics, and engineering, making them a high-priority topic for your board examinations.

In the context of your board exams, you will frequently encounter questions requiring you to solve these equations using three primary methods: factorization, completing the square, and the quadratic formula. Examiners often test your conceptual clarity by providing equations that are not in standard form, requiring you to simplify them first. Mastering these techniques ensures you can handle both simple and complex algebraic expressions with confidence.

The intuition behind quadratic equations lies in finding the roots or zeroes of a function, which represent the points where a parabola intersects the x-axis. By learning how to manipulate these equations, you are essentially learning how to solve for unknown variables in real-world scenarios, such as calculating projectile motion or optimizing areas. Consistent practice of these methods will significantly boost your marks in the algebra section of your final assessment.

2. Core Concepts and Detailed Revision Notes

2.1 Standard Form and Identification

To begin your journey with quadratic equations, you must first recognize the standard form, which is written as ax^2 + bx + c = 0. Here, x is the variable, while a, b, and c are real numbers. It is crucial to remember that 'a' cannot be zero, because if 'a' were zero, the equation would become linear rather than quadratic. When you receive a problem in your exam, your first step should always be to simplify the given expression by expanding brackets, clearing denominators, and collecting like terms on one side of the equation. This systematic approach ensures that you correctly identify the values of a, b, and c, which is the prerequisite for applying any solving method. Always check if the equation is in its simplest form before proceeding to solve it.

Key Points to Remember:

1. The standard form is ax^2 + bx + c = 0.
2. The coefficient 'a' must never be zero.
3. Always simplify the equation by moving all terms to one side.
4. Identify the values of a, b, and c carefully before calculation.
5. Quadratic equations always have a degree of two.

2.2 Solving by Factorization

Factorization is often the quickest method for solving quadratic equations when the roots are rational numbers. The process involves splitting the middle term 'bx' into two parts such that their product equals the product of 'a' and 'c', and their sum equals 'b'. Once you have split the middle term, you group the four terms into two pairs and factor out the common elements from each pair. This will result in two linear binomials being multiplied together to equal zero. According to the zero-product property, if the product of two factors is zero, then at least one of the factors must be zero. By setting each factor to zero, you can easily find the two values of x that satisfy the original equation. This method requires a strong grasp of basic multiplication tables and sign conventions.

Key Points to Remember:

1. Find two numbers that multiply to ac and add to b.
2. Split the middle term into these two numbers.
3. Group the terms and factor out common binomials.
4. Set each factor equal to zero to solve for x.
5. Verify your answers by substituting them back into the original equation.

2.3 Completing the Square

Completing the square is a powerful algebraic technique that transforms any quadratic equation into a perfect square trinomial. To perform this, you first ensure the coefficient of x^2 is one. If it is not, divide the entire equation by 'a'. Next, move the constant term to the right side of the equation. Then, take half of the coefficient of 'x', square it, and add this value to both sides of the equation. This creates a perfect square on the left side, which can be written as (x + k)^2. Finally, you take the square root of both sides, remembering to include the plus or minus sign, and solve for x. While this method is more involved than others, it is essential for understanding the derivation of the quadratic formula and is frequently required in specific board exam questions.

Key Points to Remember:

1. Ensure the coefficient of x^2 is one.
2. Move the constant term to the right side.
3. Add the square of half the x-coefficient to both sides.

4. Express the left side as a perfect square.
5. Take the square root of both sides to find x.

2.4 The Quadratic Formula

The quadratic formula is the most reliable method because it works for every quadratic equation, even those that cannot be factored easily. The formula is x = (-b plus or minus sqrt(b^2 - 4ac)) / (2a). To use it, you simply identify the coefficients a, b, and c from the standard form equation and substitute them into the formula. The expression inside the square root, b^2 - 4ac, is known as the discriminant. The value of the discriminant tells you about the nature of the roots: if it is positive, the roots are real and distinct; if it is zero, the roots are real and equal; and if it is negative, the roots are complex or imaginary. This formula is your best tool for ensuring accuracy during the examination, especially when dealing with irrational or complex roots.

Key Points to Remember:

1. The formula is x = (-b plus or minus sqrt(b^2 - 4ac)) / 2a.
2. The discriminant b^2 - 4ac determines the nature of roots.
3. This method works for all quadratic equations.
4. Be very careful with negative signs when substituting values.
5. Always simplify the final fraction to its lowest terms.

3. Key Definitions, Formulas, and Rules

1. **Standard Form:** ax^2 + bx + c = 0
2. **Quadratic Formula:** x = (-b plus or minus sqrt(b^2 - 4ac)) / 2a
3. **Discriminant:** D = b^2 - 4ac
4. **Zero Product Property:** If p-q = 0, then p = 0 or q = 0
5. **Perfect Square Trinomial:** x^2 + 2ax + a^2 = (x + a)^2
6. **Difference of Squares:** x^2 - a^2 = (x - a)(x + a)
7. **Nature of Roots (Positive):** If b^2 - 4ac > 0, roots are real and unequal
8. **Nature of Roots (Zero):** If b^2 - 4ac = 0, roots are real and equal
9. **Nature of Roots (Negative):** If b^2 - 4ac < 0, roots are imaginary

4. Step-by-Step Worked Examples

Example 1: Factorization Example

Problem: Solve x^2 - 5x + 6 = 0 by factorization.

Solution Steps:

Step 1: Identify a=1, b=-5, c=6.

Step 2: Find two numbers that multiply to 6 and add to -5. These are -2 and -3.

Step 3: Rewrite the equation as x^2 - 2x - 3x + 6 = 0.

Step 4: Factor by grouping: x(x - 2) - 3(x - 2) = 0.

Step 5: Factor out the common binomial: (x - 2)(x - 3) = 0.

Step 6: Set each factor to zero: x - 2 = 0 or x - 3 = 0.

Final Answer: x = 2, x = 3

Example 2: Quadratic Formula Example

Problem: Solve x^2 - 4x - 5 = 0 using the quadratic formula.

Solution Steps:

Step 1: Identify a=1, b=-4, c=-5.

Step 2: Substitute into formula: x = (-(-4) plus or minus sqrt((-4)^2 - 4(1)(-5))) / 2(1).

Step 3: Simplify: x = (4 plus or minus sqrt(16 + 20)) / 2.

Step 4: Simplify: x = (4 plus or minus sqrt(36)) / 2.

Step 5: Calculate: x = (4 plus or minus 6) / 2.

Step 6: Find roots: x = 10/2 = 5 and x = -2/2 = -1.

Final Answer: x = 5, x = -1

Example 3: Completing the Square Example

Problem: Solve x^2 + 6x - 7 = 0 by completing the square.

Solution Steps:

Step 1: Move constant: x^2 + 6x = 7.

Step 2: Add (6/2)^2 = 9 to both sides: x^2 + 6x + 9 = 7 + 9.

Step 3: Write as square: (x + 3)^2 = 16.

Step 4: Take square root: x + 3 = plus or minus 4.

Step 5: Solve: x = -3 + 4 = 1 and x = -3 - 4 = -7.

Final Answer: x = 1, x = -7

Example 4: Standardizing Equation

Problem: Solve x(x + 3) = 10.

Solution Steps:

Step 1: Expand: x^2 + 3x = 10.

Step 2: Move to standard form: x^2 + 3x - 10 = 0.

Step 3: Factor: (x + 5)(x - 2) = 0.

Step 4: Solve: x + 5 = 0 or x - 2 = 0.

Final Answer: x = -5, x = 2

5. Common Exam Mistakes to Avoid

1. Forgetting to set the equation to zero before identifying a, b, and c.
2. Miscalculating the discriminant by ignoring negative signs within b^2 - 4ac.
3. Failing to divide the entire equation by 'a' when completing the square.
4. Forgetting the plus or minus sign when taking the square root of both sides.
5. Incorrectly splitting the middle term during factorization due to sign errors.
6. Not checking if the final answers satisfy the original equation.

6. Practice Questions and Model Answers

Question 1

Solve 2x^2 - 7x + 3 = 0.

Answer 1: Using factorization: 2x^2 - 6x - x + 3 = 0, 2x(x - 3) - 1(x - 3) = 0, (2x - 1)(x - 3) = 0. Roots are x = 1/2, x = 3.

Question 2

Solve x^2 + 2x - 8 = 0.

Answer 2: Using factorization: (x + 4)(x - 2) = 0. Roots are x = -4, x = 2.

Question 3

Find the roots of 3x^2 + 5x - 2 = 0.

Answer 3: Using quadratic formula: a=3, b=5, c=-2. x = (-5 plus or minus sqrt(25 - 4(3)(-2))) / 6 = (-5 plus or minus 7) / 6. Roots are x = 1/3, x = -2.

Question 4

Solve x^2 - 10x + 25 = 0.

Answer 4: Factorize: (x - 5)^2 = 0. Root is x = 5 (repeated).

Question 5

Solve 4x^2 - 9 = 0.

Answer 5: Difference of squares: (2x - 3)(2x + 3) = 0. Roots are x = 3/2, x = -3/2.